HealthVision AI

AI-Based Visual Health Screening

Final Year Project Presentation

AI-Based Visual Health Screening System

A comprehensive AI-powered platform for non-invasive detection of Jaundice, Skin Diseases, and Nail Disorders using deep learning

Institution

Government Engineering College, Munger

Department of Computer Science & Engineering (AI)

Project Supervisor

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Semester

7th Semester

B.Tech CSE (Artificial Intelligence)

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Project Overview

A comprehensive AI-based health screening platform for non-invasive disease detection

Our Mission

Build a **non-invasive, AI-powered health screening platform** that uses deep learning to analyze images of eyes, face skin, and nails to detect potential health conditions — without blood tests or lab visits.

Eye Detection

Jaundice screening through sclera analysis

Face Detection

5-class skin disease classification

Nail Detection

17+ nail disorder identification

Key Statistics

Detection Modules	3
Detectable Conditions	24+
Analysis Time	<2s
Input Resolution	224×224
CNN Architectures	2

Powered by MobileNetV2 & EfficientNet Transfer Learning

Why This Project?

Addressing critical gaps in accessible health screening

Limited Rural Access

Access to **dermatological and ophthalmological screening** is severely limited in rural and underserved areas. Many patients must travel long distances just for basic health assessments, creating barriers to early detection.

Rural healthcare gap affects millions

Invasive Tests Required

Jaundice detection traditionally requires blood tests to measure bilirubin levels. Our system enables **non-invasive screening** through simple sclera image analysis — no needles, no lab visits needed.

Non-invasive alternative available

Specialist Shortage

Common skin conditions like **Acne, Eczema, Herpes, and Rosacea** require dermatological expertise. AI can assist in preliminary classification where specialists aren't available, bridging the expertise gap.

AI bridges expertise gap

Early Detection Gap

Nail abnormalities can indicate systemic health issues like liver disease, anemia, and autoimmune conditions. Early detection through image analysis prompts timely medical consultation, potentially saving lives.

Early detection saves lives

Project Objectives

The key goals we set out to achieve with HealthVision AI

Core Objectives

1. Develop a **multi-module AI platform** capable of analyzing eye, face, and nail images for disease detection
2. Implement **transfer learning** using MobileNetV2 and EfficientNetB0 for high-accuracy classification
3. Build an **intelligent eye detection pipeline** using OpenCV Haar cascades with dual-strategy detection
4. Create a **responsive web interface** supporting both image upload and real-time camera capture
5. Ensure **privacy by processing all data locally** — no images stored or transmitted to third-party services
6. Provide **detailed disease information** alongside predictions for educational and screening purposes

Key Deliverables

Three Trained Models
.h5 format for eye, face, and nail detection

Flask REST API Backend
Prediction endpoints for each module

Responsive Web Application
Camera and upload support with SPA design

Disease Information Database
Comprehensive info on 20+ conditions

Image Enhancement Pipeline
Denoising and contrast normalization

Complete Documentation
Project report and deployment instructions

How It Works

A four-step pipeline from image capture to clinical prediction

1 Capture or Upload

Use your device camera with **ROI guide overlay** or upload a photo of the eye, face, or nail for analysis.

- WebRTC camera support

2 Region Detection

OpenCV Haar cascades locate the region of interest (eye sclera, face, nail) with intelligent fallback strategies.

- Dual-strategy detection

3 AI Analysis

Detected region is **preprocessed** (224×224, normalized) and fed to the appropriate deep learning model for inference.

- < 2 seconds processing

4 Get Results

Receive **instant diagnosis**: prediction label, confidence percentage, top-k classes, and detected region preview.

- Detailed disease info

Processing Speed	Privacy	Accuracy	Device Support
< 2 Seconds	100% Local	Up to 90.62%	All Devices

Detection Module 1: Eye / Jaundice Detection

Binary classification for jaundice screening through sclera analysis

About Jaundice

Jaundice is caused by elevated bilirubin levels, which turn the sclera (white of the eye) yellow. Our model analyzes eye images to detect this discoloration for early screening.

- Non-invasive detection method
- No blood tests required
- Early warning system

Technical Specifications

Architecture **MobileNetV2**	Activation **Binary Sigmoid**
Detection **Haar Cascade**	Input Size **224×224**
Model Parameters **~2.4 Million Parameters**	

Detection Pipeline

Upload Image → Eye Detection → Sclera Crop → AI Analysis → Result

Detection Module 2: Face / Skin Disease Detection

5-class classifier for facial skin conditions using EfficientNetB0

5 Skin Conditions Detected

Condition	Severity
Acne	Mild
Eczema	Moderate
Herpes	Moderate
Panu	Mild
Rosacea	Moderate

Technical Specifications

Architecture
EfficientNetB0
ImageNet pre-trained

Classification
5-Class Softmax
Multi-class output

Normalization
BatchNorm
Stabilized training

Dense Layers
256 + Dropout 0.5
Regularization

Model Performance

Validation Accuracy **72.15%**

- Training Images: **1,196**
- Validation Images: **298**

Loss Function: **Categorical Cross-Entropy**

Detection Module 3: Nail Disease Detection

3-class nail disorder classifier for early health indicator detection

Why Nail Analysis?

Nail abnormalities can indicate systemic health issues including liver disease, anemia, fungal infections, and autoimmune conditions. Early detection through image analysis enables timely medical consultation.

- 17+ detectable conditions
- Systemic health indicators
- Non-invasive screening

3 Main Classes Detected

Healthy
Normal nail condition

Onychomycosis
Fungal infection

Psoriasis
Autoimmune condition

Technical Specifications

Architecture	Classification
MobileNetV2	3-Class Softmax

Input Size	Accuracy
224×224 pixels	85.71%

Training: **933 images** | Validation: **231 images**

Technology Stack

Built with industry-standard tools and frameworks

Deep Learning

TensorFlow / Keras
Primary ML framework

MobileNetV2
Lightweight CNN architecture

EfficientNetB0
High-efficiency model

ImageNet Transfer Learning
Pre-trained weights

Web Backend

Python 3.10+
Programming language

Flask
Web framework

Flask-CORS
Cross-origin support

REST API / JSON
Data exchange format

Computer Vision

OpenCV
Image processing

Haar Cascades
Object detection

Pillow (PIL)
Image manipulation

NumPy
Numerical computing

Frontend

HTML5
Structure

CSS3
Styling

JavaScript ES6+
Interactivity

WebRTC
Camera access

Training Datasets

Curated medical image datasets for each detection module

Jaundice Eye

ezyZip Dataset

Classes	2
Jaundice, Normal	
Training Images	136
Validation Images	32

Preprocessing

Rescale to [0, 1] (÷ 255)

Split: **80% Train / 20% Val**

Face Skin Disease

FaceZip Dataset

Classes	5
Acne, Eczema, Herpes, Panu, Rosacea	
Training Images	1,196
Validation Images	298

Preprocessing

EfficientNet preprocess_input

Split: **80% Train / 20% Val**

Nail Disease

nailZip Dataset

Classes	3
Healthy, Onychomycosis, Psoriasis	
Training Images	933
Validation Images	231

Preprocessing

Rescale to [0, 1] (÷ 255)

Split: **80% Train / 20% Val**

Data Augmentation (Training)

Rotation	Shift	Zoom	Flip	Brightness	Shear	Fill Mode
20°	15%	20%	Horizontal	0.7–1.3	10%	Nearest

Training Results & Accuracy

Two-stage training strategy with comprehensive performance metrics

Jaundice Model
MobileNetV2

Validation Accuracy
90.62%

Loss Function
Binary Cross-Entropy

Train	Val
136	32

Face Model
EfficientNetB0

Validation Accuracy
72.15%

Loss Function
Categorical Cross-Entropy

Train	Val
1,196	298

Nail Model
MobileNetV2

Validation Accuracy
85.71%

Loss Function
Categorical Cross-Entropy

Train	Val
933	231

Two-Stage Training Strategy

1. **Stage 1: Feature Extraction**
   Base frozen, LR: 1e-3, 15 epochs
2. **Stage 2: Fine-Tuning**
   Top layers unfrozen, LR: 1e-5, 25 epochs

Training Callbacks

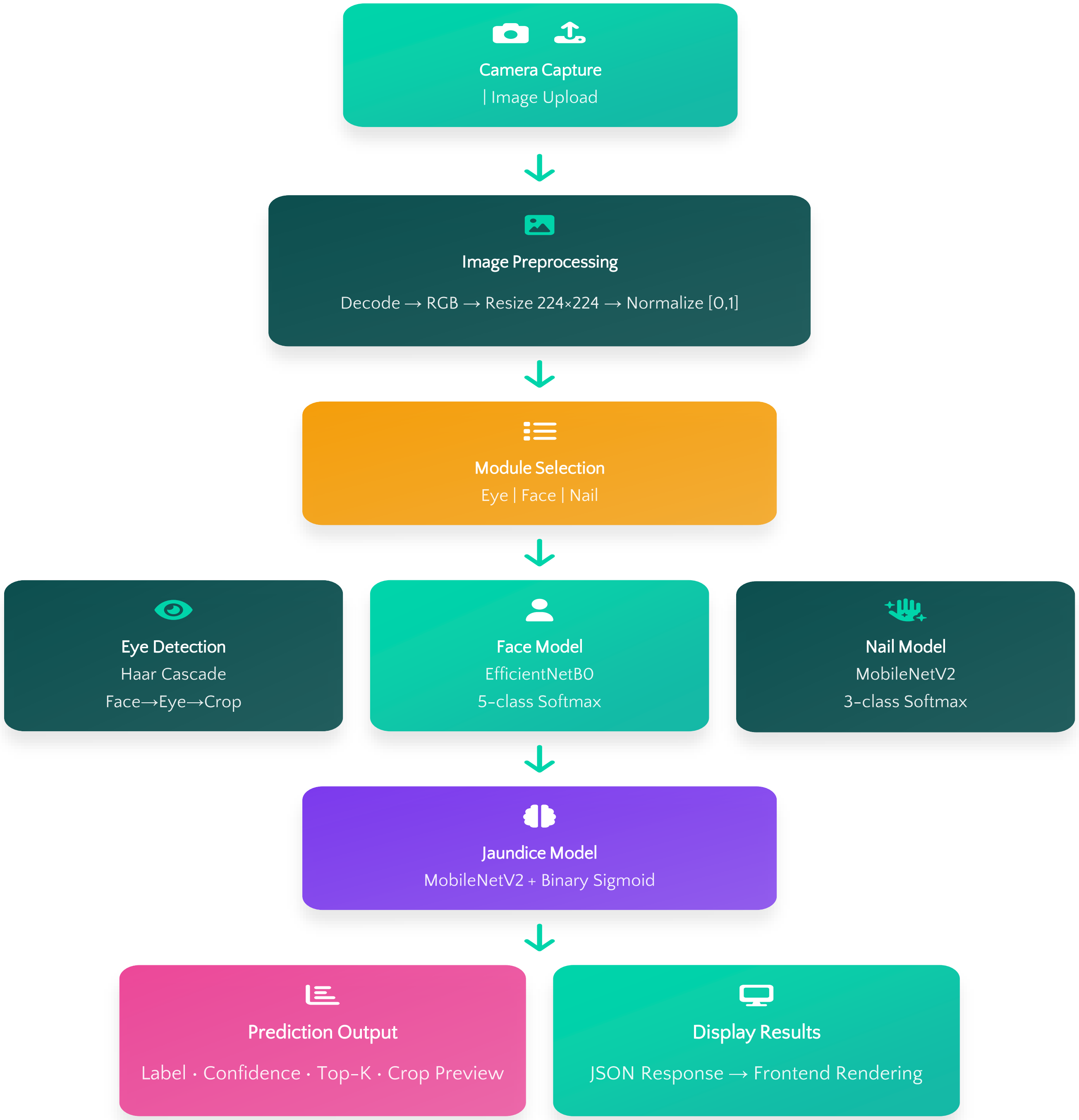
EarlyStopping
Patience: 5 epochs

ReduceLROnPlateau
Patience: 3, Factor: 0.3

ModelCheckpoint
Save best only

Application Flowchart

End-to-end data flow from user input to AI prediction output



Key Features & Capabilities

Six core capabilities that make HealthVision AI stand out

Smart Eye Detection

Haar cascade algorithm with dual-strategy detection: face-first and direct eye scanning with 35% padding for optimal sclera extraction.

- Intelligent fallback mechanism

Real-Time Camera

WebRTC camera integration with ROI guide overlay for precise image capture. Works seamlessly on both desktop and mobile browsers.

- Cross-platform compatibility

Deep Learning Models

MobileNetV2 and EfficientNetB0 transfer learning architectures pre-trained on ImageNet, fine-tuned for medical classification.

- State-of-the-art accuracy

Instant Results

Confidence scores, top-k predictions, and detected region previews generated in **under 2 seconds** per analysis.

- Lightning-fast inference

Privacy First

All processing happens **locally on the server**. No images are stored, shared, or transmitted to third-party services.

- 100% data security

Fully Responsive

Works seamlessly on **desktops, tablets, and mobile phones** with an adaptive UI that fits any screen size.

- Mobile-first design

Disease Information Database

Comprehensive coverage of 20+ detectable health conditions

Eye Conditions

2 Classes

Jaundice — Moderate

Elevated bilirubin causing yellow sclera

Normal — Healthy

Healthy eye with no discoloration

Skin Conditions

5 Classes

Acne — Mild

Clogged follicles

Eczema — Moderate

Inflammatory patches

Herpes — Moderate

Viral blisters

Panu — Mild

Fungal patches

Rosacea — Moderate

Facial redness

Nail Conditions

17+ Classes

Healthy	Onychomycosis
Psoriasis	Alopecia Areata
Beau's Lines	Bluish Nail
Clubbing	Darier's Disease
Half & Half	Koilonychia
Leukonychia	Lindsay's Nails
Muehrcke's	Onychogryphosis
Onycholysis	Pale Nail

Total Conditions
24+

Detailed Info
100%

Educational
Purpose

Thank You!

We appreciate your attention and interest in our project

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Project Team

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7th Semester CSE (AI)

Questions & Discussion

We would be happy to answer any questions about our project